

Knee Extensor and Flexor Dominant Gait Patterns Increase the Knee Frontal Plane Moment during Walking

Juha-Pekka Kulmala,^{1,2,3} Sami Äyrämö,⁴ Janne Avela¹

¹Department of Biology of Physical Activity, University of Jyväskylä, Rautpohjan Katu 8A40014 Jyväskylä, Finland, ²Tampere Research Center of Sports Medicine, UKK Institute for Health Promotion Research, Tampere, Finland, ³Department of Mechanical Engineering, Lappeenranta University of Technology, Lappeenranta, Finland, ⁴Department of Mathematical Information Technology, University of Jyväskylä, Jyväskylä, Finland

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ABSTRACT: High gait-induced knee frontal plane moment is linked with the development of knee osteoarthritis. Gait patterns across the normal population exhibit large inter-individual variabilities especially at the knee sagittal plane moment profile during loading response and terminal stance phase. However, the effects of different gait patterns on this moment remain unknown. Therefore, we examined whether different gait patterns are associated with atypically high knee frontal plane moments. Profiles of knee joint moments divided a sample of 24 subjects into three subgroups (11, 7, 6) through cluster analysis. Kinetics, kinematics, and spatio-temporal parameters were compared among clusters. Subjects who showed a typical sagittal plane moment pattern ($n = 11$) had 43% lower first peak of knee frontal plane moment compared to the cluster, which showed the dominance of the knee extensor moment during stance phase ($n = 7$, $p < 0.01$). In addition, a typical gait pattern cluster had 44% lower second peak knee frontal plane moment than the cluster, which showed the dominance of the knee flexor moment during the terminal stance phase ($n = 6$, $p < 0.05$). These findings indicate that different knee strategies driving gait considerably impact knee loading, suggesting that knee extensor and flexor dominant gait patterns demonstrate atypically high knee frontal plane moments. People in these subgroups may, therefore, be at higher risk of developing knee osteoarthritis. © 2013 Orthopaedic Research Society. Published by Wiley Periodicals, Inc. *J Orthop Res*

Keywords: gait pattern; knee moment; variability; cluster analysis; walking

Osteoarthritis (OA) is a common musculoskeletal disease causing remarkable disability for the patient and extensive societal costs.^{1,2} The medial tibiofemoral compartment of the knee is one of the most commonly affected sites.^{3,4} Although the exact mechanisms causing medial compartment OA are unclear, considerable evidence suggests that high gait-induced knee frontal plane moment (internal abduction moment), which increases medial compartment loading, plays an important role in OA initiation and progression.^{5–8} Furthermore, previous studies found that lower knee flexion excursion and higher net knee external rotation moments are also present with those who have early to moderate knee OA compared to healthy controls.^{9,10}

At the knee joint, a “typical” sagittal plane moment pattern during the loading phase of gait comprises the internal knee extensor moment driven by eccentric contraction of the quadriceps muscle. During mid-stance, the extensor moment decreases, and the moment usually changes into a flexor dominant moment for the end part of stance.¹¹ However, several studies observed a large inter-subject variability in sagittal plane knee kinetics across the normal population.^{12–14} The main variations in kinetics usually involved clear dominance of the extensor moment during the whole stance phase carried out by the greater quadriceps effort, or lower extensor moment during the loading response and a clear dominance of flexor moment

during terminal stance driven by an increased contribution of the hamstring muscles.^{12,13,15}

These findings emphasize that knee motion during the stance phase of gait can be driven with distinct kinetic strategies. However, the relationship between different knee sagittal plane kinetic patterns and the gait characteristics related to pathogenesis of knee OA are poorly understood, and to date no studies have quantified the effects of different gait patterns on the frontal plane moments. Therefore, we clustered subjects into different groups based on their knee joint moment profiles and then examined whether distinct gait patterns were associated with atypically high knee frontal moments.

METHODS

Participants

Twenty-four healthy men participated (age 40 ± 11 years, height 1.81 ± 0.05 m, and weight 83 ± 11 kg). All participants provided informed consent and confirmed no previous history of musculoskeletal problems that could affect their gait pattern. The study was approved by the local ethics committee and was performed in the accordance with the Declaration of Helsinki.

Gait Analysis

Anthropometric measurements and placement of 15 retro-reflective markers (anterior superior iliac spines, sacrum, mid-thigh, knee joint, mid-lateral leg, lateral malleoli, base of the second metatarsal and calcaneus) were used according to modified the Helen Hayes marker set.¹⁶

The subjects walked barefoot along a 10 m walkway with a self-selected velocity. Two photocells were used to control the velocity between trials ($\pm 10\%$). A five video camera system (Cohu CCD camera, San Diego, CA) and a force platform (9861A, Kistler Instrument AG, Winterthur, Switzerland)

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Correspondence to: Juha-Pekka Kulmala (T: +35-84-45718008; F: +358142602071; E-mail: juhapekka.kulmala@gmail.com)

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were used to collect marker positions and ground reaction force data synchronously at 50 and 2,000 Hz, respectively.

Kinematic and kinetic analyses were performed using the Vicon Motus motion analysis system (Oxford Metrics, UK). Marker trajectories and force data were low-pass filtered using a fourth order Butterworth filter with cut-off frequencies of 6 and 50 Hz, respectively. Right leg ground contacts were analyzed. Three trials were time normalized (0–100%) and then averaged for each subject. Identification of heel contact and toe-off events was defined based on a 20 N ground reaction force threshold. Step length and width were calculated as the difference between right and left heel marker positions at the instant of heel contact. An inverse dynamics based biomechanical model described by Vaughan et al.¹⁶ was used to calculate kinematic and kinetic data for hip, knee, and ankle joints. Joint moments were normalized to each participant's weight and are presented as internal moments to describing the muscle dominance (knee extensor vs. knee flexor) driving gait.

Data Analysis

The subjects were grouped using *K*-means algorithm on data consisting of five knee joint moment parameters (peak values during the stance phase): knee frontal plane moment (maximum abduction moment of first or second peak), knee extensor moment, knee flexor moment, and knee external rotation and internal rotation moments. Prior to data clustering, the variables were standardized by linear scaling

into the closed interval [0, 1]. The relevant dimensions were chosen by a principal component analysis (PCA) method. The first three principal components, which accounted for >90% of the total variance, were used as the input variables to the clustering algorithms. Three clustering models, for $K = \{2, 3, 4\}$, were analyzed. To avoid locally optimal clustering solutions, 100 restarts from random initial prototypes points were applied for each model. The models were validated visually through silhouette plots and comparing average silhouette indices: 0.41 ($K = 2$), 0.42 ($K = 3$), and 0.46 ($K = 4$).¹⁷ Regardless of the silhouette indices, the three-cluster model was preferred over the four-cluster model, because the group sizes in the four-cluster model were too small for further analysis. For the three-cluster model, the numbers of subjects assigned into the clusters were 6, 7, and 11. The obtained *K*-means model was evaluated by relative validation using the *K*-spatialmedians clustering method,¹⁸ which is based on non-parametric multivariate statistics that are less sensitive to outliers and deviating observations. No differences were observed between the *K*-means and *K*-spatialmedians based partitions.

The univariate between-cluster differences of selected parameters, including subject, spatio-temporal, kinetic and kinematic variables (Tables 1 and 2), were tested using nonparametric Kruskal–Wallis test. When a significant difference was found, the Mann–Whitney *U*-test with Bonferroni correction was performed to compare two clusters. $p < 0.05$ was considered significant. Results are presented as mean $\pm$ SD.

Table 1. Mean (SD) Subject, Spatio-Temporal and Kinematic Data for the Typical Gait Pattern (TYP), Knee Extensor Dominant Gait Pattern (EXT), and Knee Flexor Dominant Gait Pattern (FLX)

Parameters	Clusters			Inter-Cluster Comparisons		
	TYP (<i>n</i> = 11)	EXT (<i>n</i> = 7)	FLX (<i>n</i> = 6)	TYP versus EXT	TYP versus FLX	EXT versus FLX
Subject parameters						
Age (years)	41.5 (11.5)	34.7 (8.0)	43.0 (11.2)			
Weight (kg)	84.0 (13.3)	83.8 (5.6)	79.2 (11.0)			
Height (m)	1.80 (0.05)	1.81 (0.04)	1.82 (0.06)			
BMI (kg/m ²)	25.8 (3.2)	25.6 (1.92)	23.9 (2.6)			
Spatio-temporal parameters						
Velocity (m/s)	1.27 (0.07)	1.32 (0.08)	1.30 (0.06)			
Step length (m)	0.70 (0.03)	0.74 (0.05)	0.73 (0.02)			
Step width (m)	0.08 (0.04)	0.08 (0.01)	0.09 (0.02)			
Ankle						
Plantar flexion at HC (°)	2.1 (5.1)	−1.5 (4.4)	4.9 (5.7)			
Plantar flexion (peak) 0–50% (°)	9.3 (6.0)	5.6 (4.3)	12.6 (4.7)			0.045
Knee						
Flexion at HC (°)	5.5 (6.5)	9.3 (4.7)	0.7 (3.4)			0.009
Flexion (peak) 0–50% (°)	16.8 (5.6)	24.5 (6.3)	10.5 (3.8)	0.030		0.009
Adduction at HC (°)	1.9 (3.0)	2.0 (3.9)	1.3 (2.5)			
Adduction excursion (°)	2.8 (2.3)	2.8 (2.2)	2.3 (1.9)			
Hip						
Flexion at HC (°)	22.0 (5.7)	27.3 (6.0)	19.1 (5.1)			
Extension at TO (°)	−15.0 (8.5)	−8.5 (6.0)	−16.0 (5.3)			

Statistical analysis was done with Kruskal–Wallis test. Mann–Whitney post hoc test with Bonferroni correction was used to determine the significance between three clusters ($p < 0.05$).

Table 2. Mean (SD) Kinetic Data for the Typical Gait Pattern (TYP), Extensor Dominant Gait Pattern (EXT), and Flexor Dominant Gait Pattern (FLX)

Parameters	Clusters			Inter-Cluster Comparisons		
	TYP (<i>n</i> = 11)	EXT (<i>n</i> = 7)	FLX (<i>n</i> = 6)	TYP versus EXT	TYP versus FLX	EXT versus FLX
Ground reaction forces						
Braking	-0.19 (0.04)	-0.19 (0.04)	-0.18 (0.03)	0.012		
Propulsion	0.18 (0.02)	0.21 (0.01)	0.20 (0.01)			
Lateral	-0.03 (0.02)	-0.04 (0.03)	-0.04 (0.02)			
Medial	0.06 (0.02)	0.07 (0.02)	0.05 (0.01)			
Vertical 1	1.11 (0.10)	1.21 (0.09)	1.12 (0.08)			
Vertical 2	1.13 (0.04)	1.12 (0.04)	1.13 (0.06)			
Ankle kinetics						
Peak plantarflexor moment (Nm/kg)	1.57 (0.12)	1.50 (0.15)	1.63 (0.07)			
Knee kinetics						
Peak extensor moment (Nm/kg)	-0.58 (0.17)	-0.86 (0.11)	-0.35 (0.10)	0.012	0.015	0.009
Peak flexor moment (Nm/kg)	0.31 (0.13)	0.13 (0.18)	0.55 (0.13)		0.009	0.009
Peak abduction moment first (Nm/kg)	0.35 (0.11)	0.61 (0.16)	0.50 (0.11)	0.009		
Peak abduction moment second (Nm/kg)	0.28 (0.15)	0.45 (0.12)	0.50 (0.09)		0.021	
Peak external rotation moment (Nm/kg)	-0.12 (0.04)	-0.17 (0.03)	-0.20 (0.02)	0.024	0.003	
Peak internal rotation moment (Nm/kg)	0.03 (0.02)	0.03 (0.03)	0.04 (0.03)			
Hip kinetics						
Peak extensor moment (Nm/BW)	-0.93 (0.10)	-1.11 (0.34)	-0.97 (0.24)			
Peak flexor moment (Nm/kg)	1.17 (0.31)	1.06 (0.29)	1.27 (0.23)			
Peak abductor moment (Nm/kg)	0.89 (0.14)	1.01 (0.14)	1.11 (0.18)			

Statistical analysis was done with Kruskal–Wallis test. Mann–Whitney post hoc test with Bonferroni correction was used to determine the significance between three clusters ($p < 0.05$).

The *K*-means clustering, silhouette values and Levene's variance test were computed on MATLAB Statistics toolbox (Version 7.6, Mathworks, Natick, MA). The PCA and *K*-spatialmedians methods were implemented using standard MATLAB functions. Kruskal–Wallis and Mann–Whitney tests were performed with SPSS software (Version 18.0, SPSS, Chicago, IL).

RESULTS

Three clusters with distinct knee joint moment profiles were recognized (Fig. 1; Table 2). The largest cluster consisted of 11 subjects and was identified as the typical gait pattern (TYP). For a cluster of seven subjects, the extensor moment was apparent during almost the entire stance phase and was thus categorized as the extensor dominant gait pattern (EXT). A cluster of six subjects showed the lowest extensor moment and a clear dominance of the flexor moment during terminal stance and was thus categorized as the flexor dominant gait pattern (FLX).

No statistically significant differences among the clusters were found for subject or spatio-temporal parameters (Table 1). The propulsion ground reaction force in the TYP cluster was significantly lower than EXT clusters ($p < 0.05$, Table 2). No other differences were present in the reaction forces.

Knee sagittal plane kinetics was statistically different among three clusters. EXT produced 33% higher extensor moment compared to TYP ($p < 0.05$) and 59% higher compared to FLX ($p < 0.01$, Fig. 1; Table 2). The opposite pattern was present in the flexor moment where FLX produced 76% and 44% higher peak values than EXT ($p < 0.01$) and TYP ($p < 0.01$), respectively (Fig. 1; Table 2). In the frontal plane, TYP cluster demonstrated significantly lower knee frontal plane moment showing 43% lower First peak of knee abduction moment compared to EXT ($p < 0.01$) and 44% lower second peak of knee abduction moment compared to FLX ($p < 0.05$, Fig. 1; Table 2). In addition, consistent differences existed among clusters in the knee transversal plane kinetics, which showed significantly lower external rotation moment in the TYP compared to EXT and FLX (Fig. 1; Table 2).

Kinematics among the clusters showed a difference especially at the knee joint. The subjects in the EXT cluster demonstrated significantly higher knee flexion angle at the instant of heel contact compared to FLX ($p < 0.01$, Fig. 2; Table 1). Similarly, during loading response, EXT showed significantly higher knee flexion angle than FLX and TYP clusters ($p < 0.01$, $p < 0.05$), respectively (Fig. 2; Table 1). Similar trends in kinematic patterns were also present at the ankle and hip where

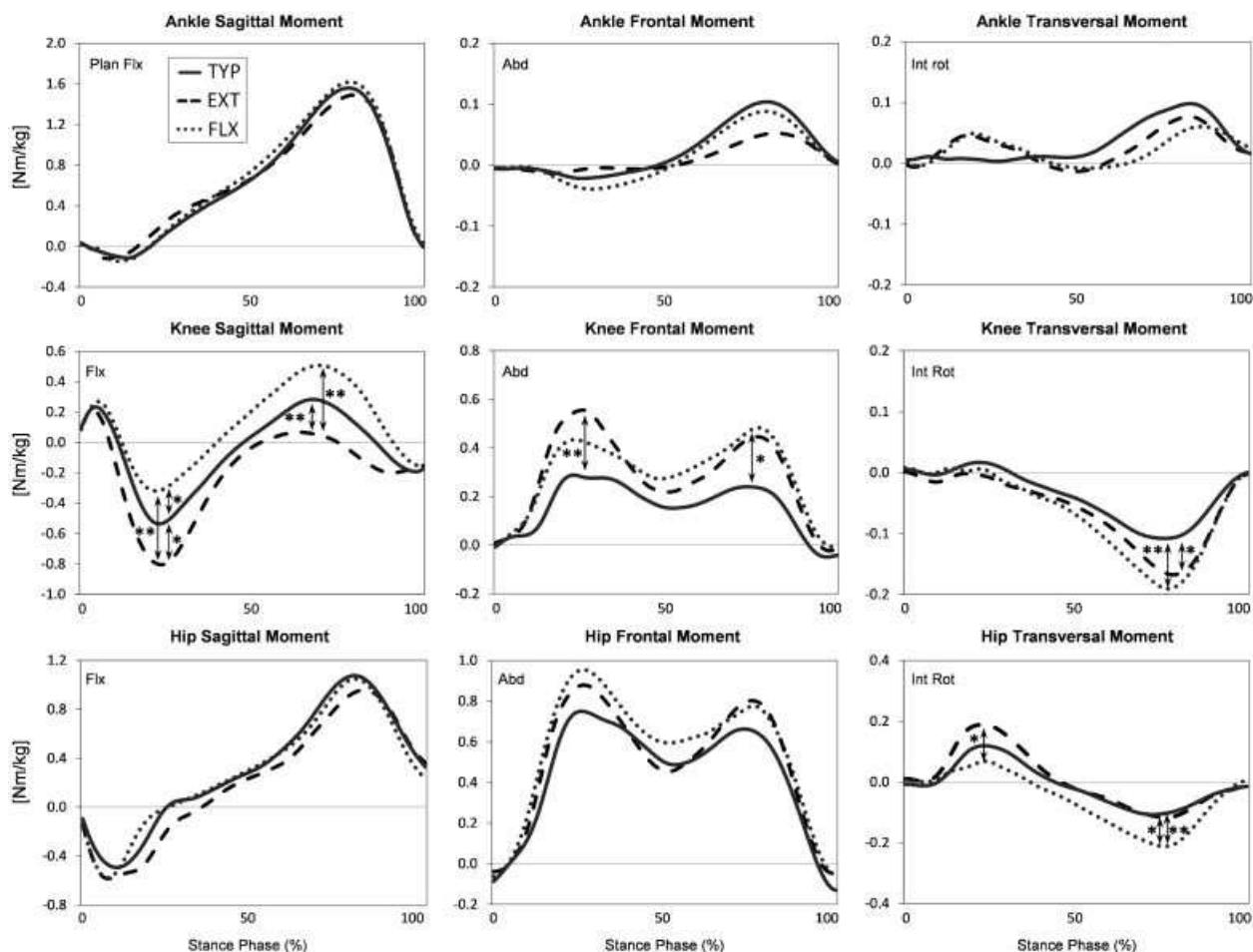


Figure 1. Mean internal joint moment curves over the stance phase of gait for the typical gait pattern (TYP), knee extensor dominant gait pattern (EXT), and knee flexor dominant gait pattern (FLX).

significantly higher peak ankle plantar flexion angle during the loading response was observed in the FLX compared to EXT ($p < 0.05$, Fig. 2; Table 1).

DISCUSSION

We examined the association between different gait patterns and knee frontal plane moment. The cluster analysis identified three clusters with distinct sagittal

plane moment patterns at the level of knee joint. This finding agrees with the previous studies^{12–14} suggesting large variability in the sagittal plane gait pattern. Furthermore, we found that distinct gait patterns demonstrated different peak knee frontal plane moment, suggesting that both extensor moment dominant (EXT) flexor moment dominant (FLX) kinetic patterns produce significantly higher frontal plane moment

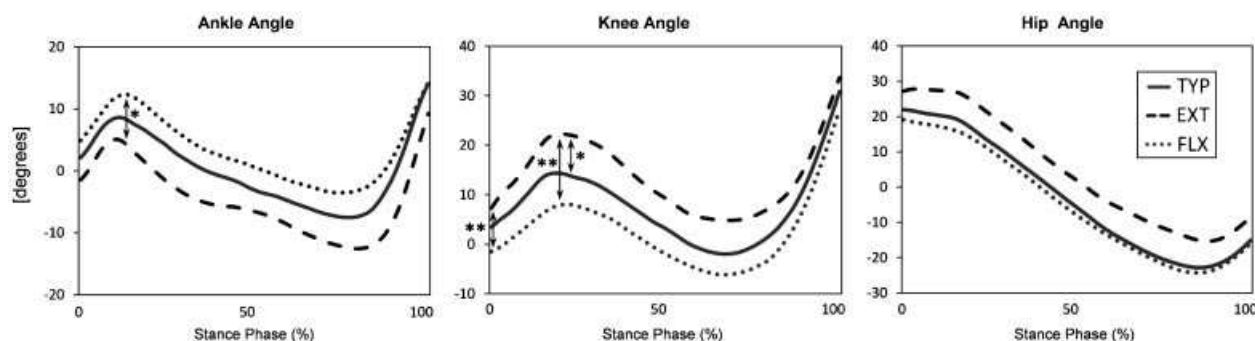


Figure 2. Mean joint angle curves over the stance phase of gait for the typical gait pattern (TYP), knee extensor dominant gait pattern (EXT), and knee flexor dominant gait pattern (FLX).

than a typical gait pattern (TYP). In addition, the TYP subjects demonstrated significantly lower knee external rotation moments than both EXT and FLX subjects.

Previous experiments utilizing the cluster analysis to study different gait patterns among healthy populations examined only sagittal plane mechanics.^{19,20} To our knowledge, the present study is the first to show the relationship between different sagittal plane gait patterns and the knee frontal plane moment during walking. High knee frontal plane moment (internal abduction moment) corresponds with the greater knee medial compression loads during walking.^{5,21} High knee loads in turn are linked to the risk of development and progression of knee OA in humans^{6,8} and animals.^{7,22} Therefore, our findings suggest that increased risk of OA may be present in those who walk with knee extensor (EXT) or flexor (FLX) dominant gait patterns. The TYP subjects also demonstrated significantly lower peak knee external rotation moments than both EXT and FLX subjects, consistent with previous studies^{9,10} suggesting the link between high frontal plane moment and increased external rotation moment of the knee. However, while the knee frontal plane moment is linked with knee OA, the role of knee rotation moment in the development of the disease remains unclear.

Because knee loads were strongly related to different gait patterns, we tested several subject, spatio-temporal, kinematic and kinetic parameters to explain underlying mechanisms responsible for these findings. There were no significant differences in spatio-temporal parameters among the clusters. A reduction of the knee extensor moment was reported after decreasing the step length,²³ and therefore we expected to see a link between different step lengths and knee extensor-flexor moment dominance. However, our results suggest that the subject-specific kinetic strategy driving the gait is independent of step length.

A recent study, where an instrumented knee implant was used to measure knee contact loads during walking, suggests that increased knee extensor moment may counteract the effects of decreased frontal plane moment and should, thereby, be considered with the frontal plane moment when gait-induced knee loads are evaluated.²⁴ In this light, the gait strategy peculiar to EXT cluster, which demonstrate both the greatest knee frontal plane moment and knee extensor moment, most likely causes the highest knee joint loads during gait.

Significant differences among clusters were observed in sagittal plane kinematics. The highest differences were present at the knee, but different kinds of profiles were present in the hip and ankle as well. The higher knee flexion during the entire stance phase was present in the EXT cluster, which most likely causes the greater knee extensor moment in this group. The opposite kinematic strategy, showing a straighter knee during the stance phase, was present in the FLX

cluster, which walked with the lowest extensor moment and highest flexor moment. A similar relationship between higher knee flexion angle and increased knee extensor moment was found by Simonsen and Alkjaer,¹⁹ who also used cluster analysis to divide subjects into groups according to gait patterns. They identified two gait patterns that showed the largest differences in the sagittal plane moment profiles and in flexion angle. Although different input parameters were used in the cluster analysis in our study, consistent findings in the sagittal plane among clusters were present. This supports the idea that variability in gait mechanics is largest at the knee joint.

The magnitude of the knee frontal plane moment is highly related to the length of the ground reaction force vector lever arm at the knee in the frontal plane.²⁵ Thus, in addition to kinematic differences, also the differences in the medio-lateral reaction forces could contribute to the variability of the frontal plane moment. However, only the horizontal propulsion force was found to be significantly higher in the EXT compared to the TYP cluster. Findings of previous studies also suggest that the magnitude of the knee frontal plane moment may be related to altered ankle plantarflexor^{26,27} or hip abductor contribution during gait.^{28,29} However, the lack of significant inter-cluster differences in the reaction forces and ankle and hip moments in our study suggests that the knee sagittal plane movement pattern may be more important in explaining the variability in the knee frontal plane moment.

Conservative treatment strategies such as muscle strengthening and laterally wedged shoes are commonly used in the management of knee OA.³⁰ However, recent studies suggest that the changes in gait mechanics show inconsistent response to these strategies,^{31,32} and some people actually demonstrate adverse biomechanical changes leading to increased knee loading.³³ Besides anatomical differences, variability in gait mechanics may also play a role.³¹ Our findings may, therefore, offer a new insight into clinical trials that have failed to demonstrate a consistent impact of conservative strategies for knee OA. Future studies examining the effects of different gait patterns on the knee adduction moment together with other factors (e.g., hip abductor strength, hip/upper body movement, and muscle co-contraction) to knee loads are required to better understand the relationship between different gait patterns and knee joint loading.

Our study has several limitations. First, a relatively small sample size possibly decreased the number of clusters that could be identified using the *K*-means analysis method. Also a self-selected gait velocity may be a limitation. However, we decided to allow subjects to use a self-selected velocity to obtain the best representation of their natural gait patterns. In addition, no differences in the mean velocity among

clusters suggest that the inter-subject differences in walking mechanics are not due to different gait velocities. Finally, we did not determine the frontal knee alignment. Since, various alignment is generally linked to higher frontal plane moment, it is an important factor to take into account. However, the dynamic knee adduction angle, which relates directly with static alignment,³⁴ showed neutral to moderate varus-alignment (range 0.4°–8.8°) in our subjects. The lack of difference in the dynamic knee adduction suggests that similar knee alignment was present among the clusters.

In summary, we identified three clusters of subjects with distinct knee joint moment patterns. Subjects who demonstrated knee extensor or flexor dominant gait patterns exhibited a significantly higher knee frontal plane moment during gait compared to those who had a typical gait pattern. Our findings suggest that individual knee strategies driving gait have considerable impact on knee frontal plane moment. Therefore, people demonstrating either knee extensor or flexor dominant gait patterns may be at higher risk of developing knee OA.

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